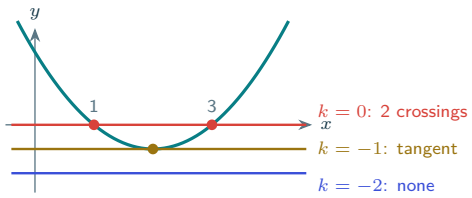
ALGEBRA ON ONE PAGE

Solving intersections: lines meet curves

Every quadratic is a line crossing a curve, and the discriminant decides how and what kind of number results.

1 / THE RULE AND THE PICTURE

Set the curve equal to the line and the crossings are the roots



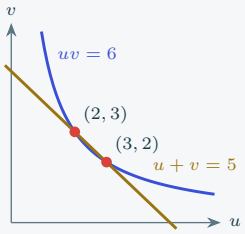
The parabola $y = x^2 - 4x + 3$ meets the line $y = k$ where the two y-values agree. For $y = x^2 + bx + c$ this reads

$$x^2 + bx + (c - k) = 0, \qquad \Delta = b^2 - 4(c - k)$$

so raising the line by 1 raises Δ by 4. Here $\Delta = 4 + 4k$: two crossings at $k = 0$ (roots $1, 3$), one repeated at the vertex $k = -1$ (root 2) and none at $k = -2$.

2 / VIETA AND THE DISCRIMINANT / SUM, PRODUCT AND SEPARATION

Roots as a point where a line meets a hyperbola, and Δ as a distance test

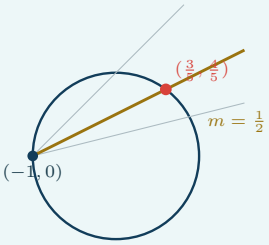


If the roots of $x^2 + Bx + C = 0$ are x_1, x_2 then $x_1 + x_2 = -B$ and $x_1 x_2 = C$, so with $s = -B$ and $p = C$ the roots are the coordinates where the line $u + v = s$ meets the hyperbola $uv = p$, at (x_1, x_2) and (x_2, x_1); $s = 5$, $p = 6$ gives 2 and 3, and a line that misses the hyperbola means complex roots. For $ax^2 + bx + c$ the roots sit at $-b/2a \pm \sqrt{\Delta}/2a$, so Δ measures their separation:

$\Delta > 0$	two real crossings, a perfect square giving rational roots
$\Delta = 0$	one repeated root, the line is tangent
$\Delta < 0$	no real crossing, complex conjugate roots

3 / A KNOWN ROOT / THE CIRCLE COLLAPSES THE QUADRATIC

Lines through $(-1, 0)$ meet the unit circle again at a rational point



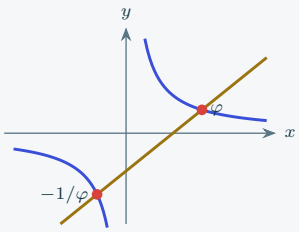
Substituting $y = m(x + 1)$ into $x^2 + y^2 = 1$ gives a quadratic that must contain the factor $(x + 1)$, because $(-1, 0)$ is already on the circle. Dividing it out leaves a linear equation with rational coefficients, so

$$x = \frac{1 - m^2}{1 + m^2}, \qquad y = \frac{2m}{1 + m^2}$$

and $m = \frac{1}{2}$ gives $(\frac{3}{5}, \frac{4}{5})$, the Pythagorean triple $3, 4, 5$ once the denominators are cleared.

4 / NO KNOWN ROOT / THE HYPERBOLA KEEPS THE QUADRATIC

The line $y = x - k$ meets $xy = 1$ where $x^2 - kx - 1 = 0$



No root is known in advance, so the full quadratic survives, $x = \frac{k}{2} \pm \frac{1}{2}\sqrt{k^2 + 4}$. Since $k^2 < k^2 + 4 < (k + 1)^2$ for $k \geq 2$ and 5 is not a square, $\Delta = k^2 + 4$ is never a perfect square and the roots are always irrational:

k	1	2	3	4
$\Delta = k^2 + 4$	5	8	13	20
positive root	golden φ	silver $1 + \sqrt{2}$	bronze $\frac{3+\sqrt{13}}{2}$	$2 + \sqrt{5}$

5 / SUMMARY / WHAT THE GRAPH IS SAYING

A known root gives rational points; no known root gives metallic means

construction	equation	Δ	roots	meaning
parabola at $y = 0$	$x^2 - 4x + 3 = 0$	4	$1, 3$	two rational crossings
same parabola at $y = -1$	$x^2 - 4x + 4 = 0$	0	2 repeated	line tangent at the vertex
circle, $m = \frac{1}{2}$	root -1, then $\frac{3}{5}$	linear	rational	the point $(\frac{3}{5}, \frac{4}{5})$
hyperbola, $k = 1$	$x^2 - x - 1 = 0$	5	φ, $-1/\varphi$	golden mean and conjugate
hyperbola, $k = 2$	$x^2 - 2x - 1 = 0$	8	$1 \pm \sqrt{2}$	silver mean and conjugate
Vieta, $s = 5$, $p = 6$	$x^2 - 5x + 6 = 0$	1	$2, 3$	$u + v = 5$ meets $uv = 6$

ASIDE / MONIC FORM

Any $ax^2 + bx + c$ divides by a to become monic, so taking $a = 1$ loses nothing and the leading coefficient leaves every formula.

ASIDE / CHECK IN PYTHON

`np.roots([1,-1,-1])` gives $1.618, -0.618$, and `(1-m**2)/(1+m**2)` at `m=0.5` gives 0.6.

Convention. Roots are where the curve and line share a point; $\Delta = b^2 - 4ac$. After the Lines Meet Curves preamble.